

Simulating aquaculture breeding

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Agenda

A model of the Swedish rainbow trout breeding program based on Kurta et al. (2023).

The most complex breeding structure we've seen so far — with custom cross plan.

The study

Kurta K, Jeuthe H, de Koning D-J, Palaiokostas C. Insights from the early generations of the Swedish rainbow trout (*Oncorhynchus mykiss*) breeding program. Acta Agriculturae Scandinavica, Section A — Animal Science. 2023;72: 88–99. doi:[10.1080/09064702.2023.2245490](https://doi.org/10.1080/09064702.2023.2245490)

Breeding program (re-)established 2011
2 generations (2016, 2019)

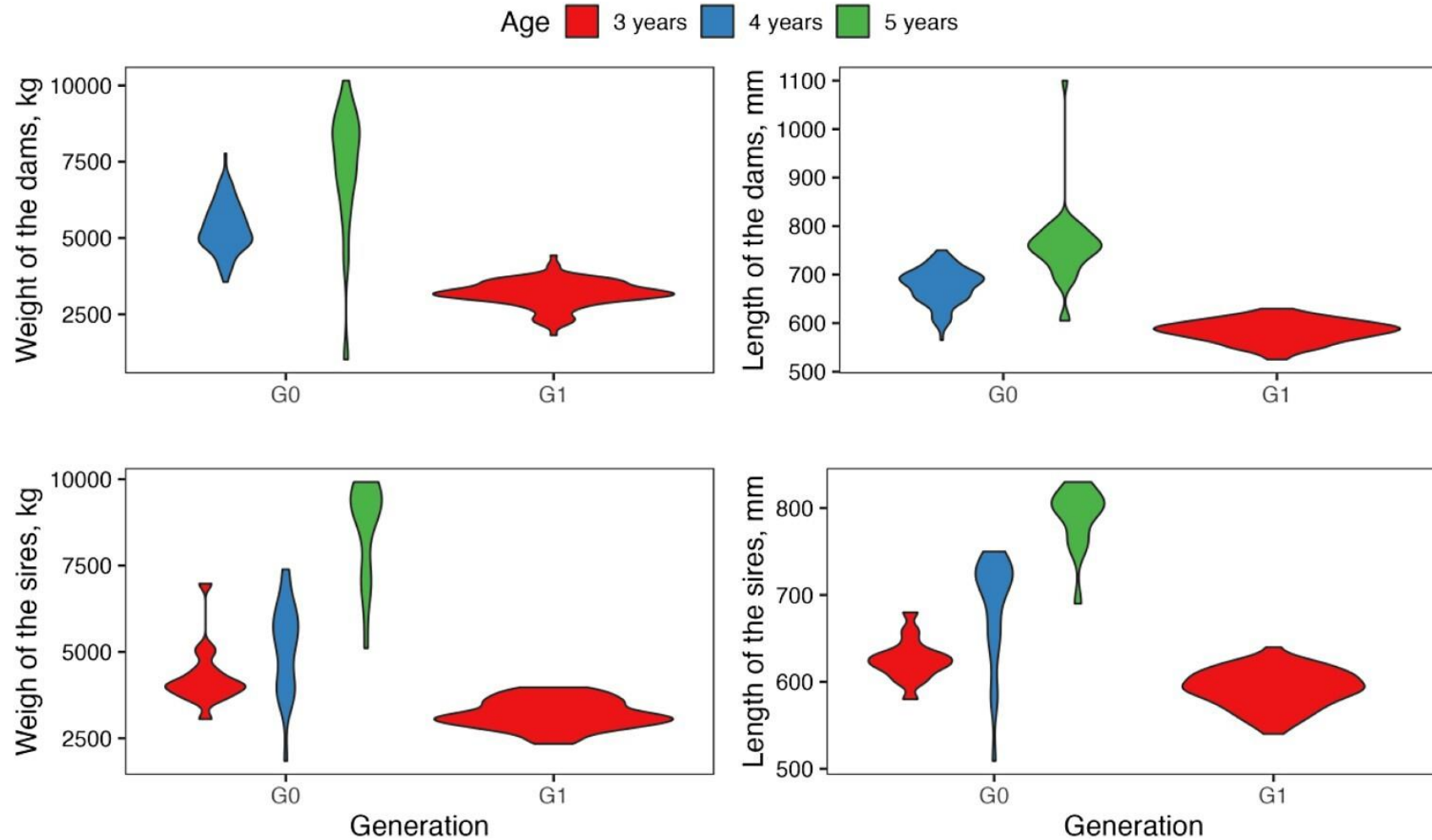
What are the genetic parameters and response to selection?

Family-based breeding program

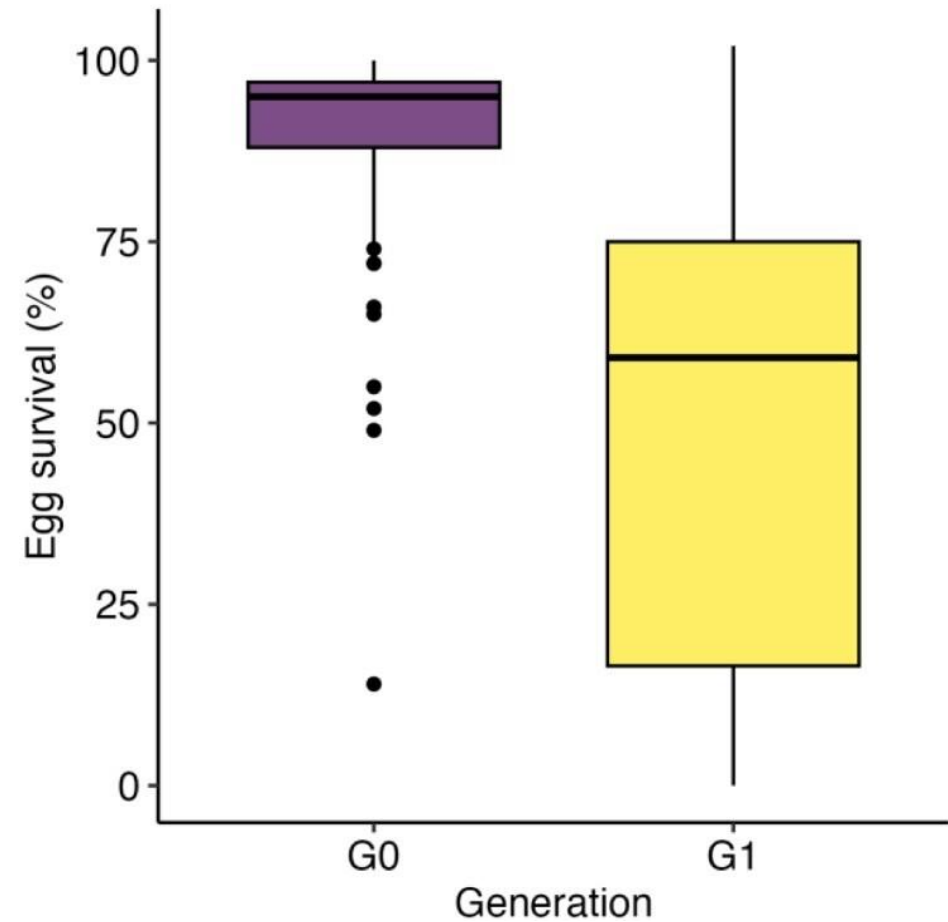
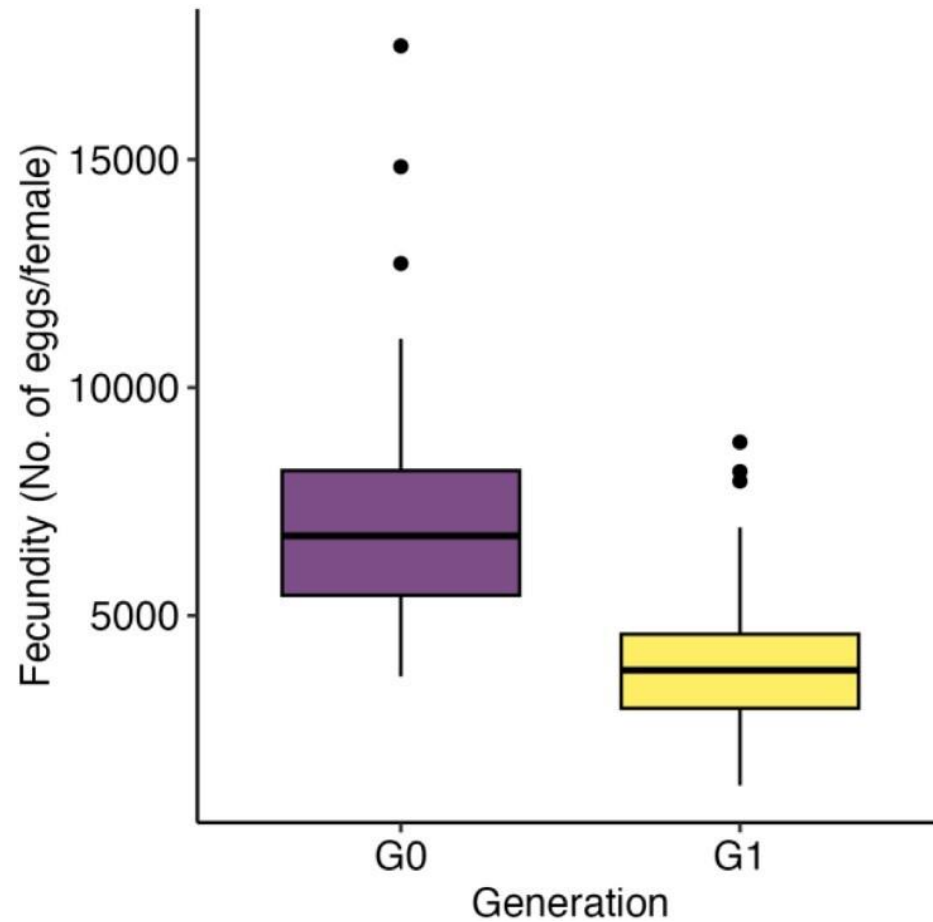
Nested breeding design, 2 females fertilised with 1 male

”Relatively small” for salmonid breeding program

Generation	No. of crosses	No. of crosses with viable offspring	Sire	Dam	Offspring
G0	133	133	67	133	1768
G1	116	73	58	116	1643
Overall	249	206	125	249	3411

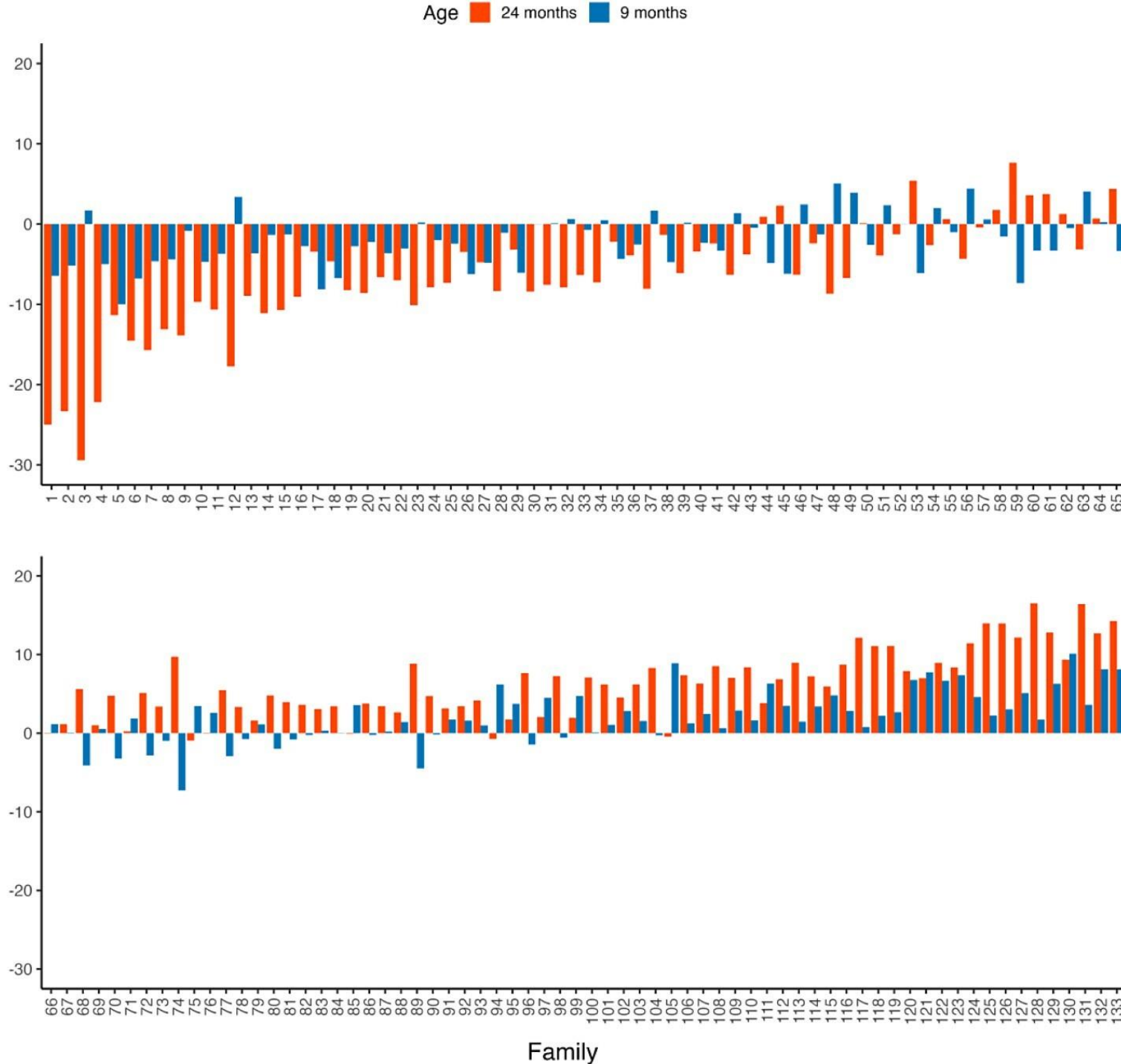


Traits recorded: body weight and body length at different ages.



Not genetically analysed: reproductive traits. Substantial differences between generations, but also measured at different ages.

Estimated breeding value of G0



EBVs for body weight at two ages.

Selection on growth, as well as independent culling of deformations and early maturation of males.

Trait	Length 1st (SE)	Weight 1st (SE)	Length 2nd (SE)	Weight 2nd (SE)
Length 1st (SE)	0.39 ^(0.16)	0.95 ^(0.04)	0.23 ^(0.56)	0.27 ^(0.03)
Weight 1st (SE)	0.95	0.32 ^(0.15)	0.19 ^(1.03)	0.48 ^(0.38)
Length 2nd (SE)	0.34	0.31	0.23 ^(0.11)	0.94 ^(0.01)
Weight 2nd (SE)	0.34	0.35	0.94	0.37 ^(0.12)

Genetic parameters for the four traits in G0. Genetic correlations above diagonal, phenotypic below.

The simulation

All details are not reported (exact family sizes, precise selection method etc), but that is okay.

We're not trying to make an exact digital twin of the breeding program, only something that has roughly similar properties and that can be used to generate replicates.

Goal:

- Illustrate custom cross plans in AlphaSimR
- Implement family selection and within-family selection

Aquaculture populations were domesticated (captured) much more recently than terrestrial livestock. The GENERIC population history is not a good fit.

The 2011 rainbow trout program was founded from "remnants of the original strain in combination with broodstock from the Swedish rainbow trout industry".

```
founders <- runMacs2(nInd = 300,  
                    nChr = 10,  
                    Ne = 1000,  
                    histNe = NULL,  
                    histGen = NULL)
```

```
simparam <- SimParam$new(founders)
simparam$setSexes("yes_sys")

effect_correlation_matrix <- matrix(
  c(1, 0.23,
    0.23, 1),
  ncol = 2,
  byrow = TRUE
)

simparam$addTraitA(nQtlPerChr = 100,
  mean = c(160, 500),
  var = c(15^2, 30),
  corA = effect_correlation_matrix)

simparam$setVarE(h2 = c(0.39, 0.23))
```

```
founderpop <- newPop(founders, simParam = simparam)
```

```
n_sires <- c(67, 58)
```

```
n_dams <- c(133, 116)
```

```
n_viable_crosses <- c(133, 73)
```

```
n_offspring <- c(1768, 1643)
```

```
generations <- vector(mode = "list", length = 2)
```

Custom cross plans

The `makeCross`/`makeCross2` functions pair animals according to a "cross plan", which is just a matrix of IDs of parents

`randCross`, `randCross2`, use `makeCross` under the hood

Custom cross plans are useful for complex designs, optimal contributions, etc.

Here, we write a function to create the right number of families and offspring each generation.

First some parameters and book keeping.

```
make_cross_plan <- function(sires, dams, n_viable_crosses, n_offspring) {  
  n_dams <- dams@nInd  
  n_sires <- sires@nInd  
  offspring_per_family <- n_offspring/n_viable_crosses  
  
  families <- matrix("", nrow = n_dams * 2, ncol = 2)
```

The final cross plan has one row per offspring. We start with one per family.

Generate all families that are attempted; we will remove non-viable families later.

```
## Create families where each dam is used twice, and sires are used
## at random, but at least once
families[, 1] <- rep(dams@id, each = 2)
families[1:n_sires, 2] <- sample(sires@id)
n_extra_sires <- nrow(families) - n_sires
families[(n_sires + 1):nrow(families), 2] <-
  sample(sires@id, n_extra_sires, replace = TRUE)
```

Randomly select the families where crossing was successful.

```
## Sample successful crossings  
successful_families <- families[sample(1:nrow(families), n_viable_crosses),]
```

Calculate how many offspring per family we need to generate to get the desired total number of offspring.

```
## Calculate offspring per family; first make equal family sizes,  
## then randomly apportion the extra offspring  
n_per_family <- rep(floor(offspring_per_family), each = nrow(successful_families))  
n_extra_offspring <- n_offspring - sum(n_per_family)  
offspring_to_add <- sample(1:length(n_per_family), n_extra_offspring)  
for (extra in 1:length(offspring_to_add)) {  
  n_per_family[offspring_to_add[extra]] <- n_per_family[offspring_to_add[extra]] + 1  
}
```

Finalise the cross plan by replicating each family row for the number of family members.

And return the plan at the end of the function

```
## Expand the mating plan by replicating each row the right number of times
cross_plan_list <- vector(mode = "list", length = nrow(successful_families))
for (family in 1:length(cross_plan_list)) {
  cross_plan_list[[family]] <- matrix(rep(successful_families[family, ],
                                          n_per_family[family]),
                                     ncol = 2, byrow = TRUE)
}
cross_plan <- Reduce(rbind, cross_plan_list)
cross_plan
}
```

```
## G0: no selection
```

```
sires <- selectInd(founderpop,  
                   sex = "M",  
                   method = "rand",  
                   nInd = n_sires[1],  
                   simParam = simparam)
```

```
dams <- selectInd(founderpop,  
                  sex = "F",  
                  method = "rand",  
                  nInd = n_dams[1],  
                  simParam = simparam)
```

```
cross_plan <- make_cross_plan(sires, dams, n_viable_crosses[1], n_offspring[1])

generations[[1]] <- makeCross2(females = dams,
                                males = sires,
                                crossPlan = cross_plan,
                                simParam = simparam)
```

Family-based selection

- First select the top families based on family means, `selectFam`
- Then select the best candidate within selected families based on their individual record, `selectWithinFam`
- Exact procedure not reported; we use 1:1 index

We use phenotypic selection with individual records, but many options are possible.

- Family-based selection on traits that are destructively measured (e.g., slaughter records)
- Pedigree BLUP and genomic prediction

```
## G1: Family-based selective breeding
```

```
candidate_sire_families <- selectFam(generations[[1]],  
                                     sex = "M",  
                                     trait = selIndex,  
                                     b = c(1, 1),  
                                     nFam = n_sires[2],  
                                     simParam = simparam)
```

```
sires <- selectWithinFam(candidate_sire_families,  
                        trait = selIndex,  
                        b = c(1, 1),  
                        nInd = 1,  
                        simParam = simparam)
```

```
candidate_dam_families <- selectFam(generations[[1]],  
                                     sex = "F",  
                                     trait = selIndex,  
                                     b = c(1, 1),  
                                     nFam = n_dams[2],  
                                     simParam = simparam)
```

```
dams <- selectWithinFam(candidate_dam_families,  
                        trait = selIndex,  
                        b = c(1, 1),  
                        nInd = 1,  
                        simParam = simparam)
```

```
cross_plan <- make_cross_plan(sires, dams, n_viable_crosses[2], n_offspring[2])
```

```
generations[[2]] <- makeCross2(females = dams,  
                                males = sires,  
                                crossPlan = cross_plan,  
                                simParam = simparam)
```

Custom cross plans and makeCross can be used for interesting breeding strategies.

Family selection and within-family selection combined in aquaculture.

Population history of aquaculture species different from terrestrial farm animals.

What could we use this model to learn about?

- Effect of family size and family number on genetic gain and variation?
- Potential for genomic prediction in this breeding design?
- Dealing with gene-by-sex interactions?
- Expected genetic gain compared to equation-based prediction?

Image credits

Kurta K, Jeuthe H, de Koning D-J, Palaiookostas C. Insights from the early generations of the Swedish rainbow trout (*Oncorhynchus mykiss*) breeding program. *Acta Agriculturae Scandinavica, Section A — Animal Science*. 2023;72: 88–99. doi:[10.1080/09064702.2023.2245490](https://doi.org/10.1080/09064702.2023.2245490)
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